Benchmarking Deep-Learning and Waveform-Migration Seismic Detection Tools for Automated Earthquake Catalogue Generation in the Taupō Volcanic Zone, New Zealand

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Abstract

Automated earthquake detection tools now underpin the construction of modern seismic catalogues, but their relative performance varies between regions, and no published study has compared deep-learning phase pickers directly against a waveform-migration method within the Taupō Volcanic Zone (TVZ), New Zealand. This project benchmarked three such methods, EQTransformer with PyOcto, PhaseNet with GaMMA, and QuakeMigrate, against one another and against an independent reference catalogue, over a two-month window (August 1st to September 30th 2019) encompassing a documented volcanic unrest period and a magnitude 5.2 mainshock earthquake. All three pipelines processed the same waveform archive from 60 GeoNet stations, and shared identical absolute location, quality filtering and local magnitude methods, so that differences between the final catalogues are attributable to the detection stage.

After quality filtering, EQTransformer retained 904 events, PhaseNet 933, and QuakeMigrate 31. Against the reference catalogue, EQTransformer achieved 61.2% precision and 53.7% recall, PhaseNet 57.1% and 51.6%, and QuakeMigrate 83.9% and 2.5%. Magnitudes of completeness, estimated by maximum curvature, were 1.55, 1.45 and 1.95 respectively, all substantially lower than the routine national catalogue's approximate threshold of 2.0 in this region. Gutenberg-Richter b-values ranged from 0.729 ± 0.177 (QuakeMigrate) to 1.594 ± 0.103 (PhaseNet).

Three findings recurred independently across all three catalogues: events unmatched to the reference catalogue were systematically deeper than matched events; azimuthal gap was the dominant quality-filter rejection criterion; and seismicity concentrated sharply around the September 4th mainshock. That structurally different tools converge on the same signatures indicates these reflect shared limitations of the velocity model and station network geometry rather than any single algorithm. The spread in estimated b-values is sufficient to shift an extrapolated large-event exceedance rate for the TVZ by a policy-relevant margin, with direct implications for the seismicity-rate component of national seismic hazard and catastrophe models.

List of Contents

Title Page **1**

Abstract **2**

List of Contents **3**

Acknowledgements **5**

1. Introduction **6**

1.1 Background and Context 6

1.2 The Research Gap 7

1.3 Applied and Industry Motivation 8

1.4 Aims, Objectives and Research Questions 8

1.5 Terminology 9

1.6 Study Region 10

2. Methodology **10**

2.1 Study Area, Time Window and Data 10

2.2 Comparative Framework 11

2.3 Phase Detection and Association 11

2.3.1 Phase 1: EQTransformer and PyOcto 12

2.3.2 Phase 2: PhaseNet and GaMMA 12

2.4 Absolute Location 13

2.5 Quality Filtering 13

2.6 Relative Relocation 14

2.7 Local Magnitude Estimation 14

2.8 Phase 3: QuakeMigrate 15

2.8.1 Travel-Time Lookup Table 15

2.8.2 Detection and Event Triggering 15

2.8.3 Location and Local Magnitude Estimation 16

2.8.4 Quality Filtering 16

2.9 Comparative Analysis Methods 17

3. Results **18**

3.1 Catalogue Overview 18

3.2 Benchmark Comparison 19

3.3 Magnitude of Completeness and b-value 19
3.4 Spatial and Depth Distribution 20
3.5 Cross-Catalogue Agreement 20
3.6 Sub-Region Distribution and Magnitude-Frequency 20
3.7 Temporal Distribution 21
3.8 Seismicity Maps 22
3.9 Seismic Density Mapping 24
3.10 Depth-Stratified and Magnitude-Filtered Density 26
3.11 Catalogue Difference Maps 27
3.12 Extended Temporal Analysis 28
3.13 Depth Cross-Sections 29
4. Discussion 30
4.1 Reproducibility of Illsley-Kemp and Mestel (2025) 30
4.2 Convergent Methodological Findings 31
4.3 Precision/Recall Trade-offs Across Tools 32
4.4 The QuakeMigrate Threshold-Selection Story 33
4.5 The August/September Activity Pattern 34
4.6 Implications for Hazard Characterisation and Catastrophe Modelling 35
4.7 Limitations 37
5. Conclusions 38
References 40

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1. Introduction

1.1 Background and Context

The Taupō Volcanic Zone (TVZ) is one of the most seismically and volcanically active rift systems in the world, accommodating oblique extension between the Pacific and Australian plates across the central North Island of New Zealand. Earthquake activity in the TVZ is closely tied to its volcanic and geothermal systems, with rift-related faulting, magmatic intrusion and hydrothermal processes all contributing to the region's seismicity. Characterising this seismicity accurately, down to the smallest reliably detectable magnitudes, matters directly for two related reasons: small earthquakes constrain the Gutenberg-Richter b-value and magnitude-frequency relationship used throughout seismic hazard analysis; and elevated small-earthquake rates are a primary observable signal of volcanic unrest (Wiemer and Wyss, 2000).

GeoNet, New Zealand's national geohazard monitoring network, provides the country's routine earthquake catalogue, but its real-time detection threshold in the TVZ is limited to a magnitude of completeness (Mc) of approximately 2.0 (Illsley-Kemp and Mestel, 2025). This limitation is well recognised, and has motivated dedicated research efforts to build more complete regional catalogues using automated, non-routine detection methods. The most direct such effort, and the direct predecessor this project reproduces and extends, is Illsley-Kemp and Mestel (2025), who applied EQTransformer for phase picking, PyOcto for phase association, NonLinLoc for absolute location and GrowClust3D for relative relocation to sixteen years of TVZ seismicity (2007-2023), achieving a magnitude of completeness of approximately 1.2, more than half a magnitude unit lower than GeoNet's routine catalogue.

This project sits within a broader, rapidly developing field. The application of deep-learning phase pickers to automated earthquake catalogue generation, which has moved from a niche research topic to a widely adopted operational tool within roughly a decade (Mousavi and Beroza, 2022). PhaseNet (Zhu and Beroza, 2019) and EQTransformer (Mousavi et al., 2020), the two deep-learning pickers used in this project, are among the most widely applied models in this field, and both were developed as general-purpose pickers intended to generalise across regions and networks beyond the specific data they were trained on.

That intended generalisation has not always held in practice. Münchmeyer et al. (2022) conducted a systematic, quantitative evaluation of six deep-learning pickers, including both PhaseNet and EQTransformer, across eight independent datasets, and found meaningful performance differences between models that depended on the specific dataset used, establishing direct, regional benchmarking of this kind as a an established research activity

rather than a purely academic exercise. More specifically, Jiang et al. (2021) compared PhaseNet and EQTransformer directly against each other on two earthquake sequences in China (the 2021 Yangbi and Maduo earthquakes), finding that PhaseNet achieved higher recall than EQTransformer in that setting, and that both models' recall fell by 13-56% relative to the performance reported in their own original papers once applied outside their original training data. This is a useful, cautionary data point for the present project specifically. It is the opposite precision/recall ordering to the one found here (Section 3.2), where EQTransformer achieved a marginally higher recall than PhaseNet. Taken together, these findings indicate that the relative performance of these tools is not fixed, but depends on the specific region and dataset to which they are applied, which is itself a direct motivation for undertaking a TVZ-specific comparison rather than assuming that findings from elsewhere transfer directly.

QuakeMigrate (Winder et al., 2026), the third detection method used in this project, represents a structurally different approach to the same underlying problem. Rather than picking phase arrivals at individual stations before associating them into events, QuakeMigrate works by migrating and stacking waveform energy across the whole network simultaneously, searching for the spatial and temporal coalescence of energy that indicates a real earthquake source. This network-level approach does not depend on any individual station achieving a high signal-to-noise ratio, which in principle makes it well suited to noisy, geothermally active settings such as the TVZ.

1.2 The Research Gap

No prior study has directly compared EQTransformer, PhaseNet and QuakeMigrate against one another specifically within the TVZ. Extensive literature exists comparing deep-learning pickers such as EQTransformer and PhaseNet against one another in various settings (Section 1.1), but no published study was found that directly benchmarks a waveform-migration method such as QuakeMigrate against deep-learning pickers of this kind, in the TVZ or elsewhere.

This gap matters because the three methods embody fundamentally different detection philosophies, each with different, well-documented strengths and weaknesses. EQTransformer and PhaseNet are both picker-then-associate methods, they first identify candidate phase arrivals at individual stations, then associate those picks across the network into candidate events. This approach is computationally efficient and scales well to long time windows, but its performance is bounded by how reliably phases can be picked at each station individually, particularly for small, low signal-to-noise events. QuakeMigrate instead searches directly for coherent energy across the whole network, which can recover events too weak to

produce a confident single-station pick, at the cost of being substantially more computationally expensive and requiring careful, manually chosen detection thresholds (Section 2.8).

The TVZ presents a demanding test case for comparing these approaches directly. Its heterogeneous, actively deforming crust and locally variable seismic velocity structure complicate both phase picking and event location, strong volcanic path attenuation reduces signal amplitude with distance in ways not well captured by standard regional ground-motion assumptions, and persistent geothermal and volcanic noise reduces the signal-to-noise ratio available to any single-station picker. A method's performance under these specific conditions cannot be safely assumed from its performance in a different tectonic and noise setting, which is precisely what motivates undertaking this comparison directly in the TVZ rather than relying on results reported elsewhere.

1.3 Applied and Industry Motivation

Beyond its academic interest, this project has a direct applied motivation. New Zealand's earthquake insurance and reinsurance market, shaped substantially by the experience of the Canterbury earthquake sequence and the ongoing role of the Earthquake Commission (EQC), depends on catastrophe models such as Moody's RMS and Verisk AIR Worldwide's New Zealand earthquake models to price and manage seismic risk. These models depend, in turn, on regional seismicity-rate characterisation, commonly informed by exactly the kind of magnitude-of-completeness and Gutenberg-Richter b-value estimates this project produces for the TVZ. The specific, concrete implications of this project's results for that applied context, including a direct connection to the seismicity-rate component of the 2022 New Zealand National Seismic Hazard Model that Moody's RMS explicitly incorporates, are discussed in full in Section 4.6, once the relevant results have been presented.

1.4 Aims, Objectives and Research Questions

This project aims to benchmark three automated earthquake detection and location methods, EQTransformer/PyOcto, PhaseNet/GaMMA and QuakeMigrate, against one another and against an independent benchmark catalogue, within the TVZ. It is structured around three research questions:

RQ1: Can the workflow of Illsley-Kemp and Mestel (2025) be reproduced using EQTransformer and PyOcto, and to what extent does a reproduction of this kind match the original study's own reported performance?

RQ2: How do PhaseNet/GaMMA and QuakeMigrate compare to EQTransformer/PyOcto in terms of catalogue completeness, recall, hypocentral location precision and computational efficiency, within the TVZ specifically?

RQ3: What are the implications of these results for hazard characterisation and catastrophe modelling in the TVZ?

The project's originally proposed study period spanned seventeen years (2007-2023), matching the Illsley-Kemp and Mestel (2025) benchmark catalogue's own full duration. For reasons explained in full in the Methodology (Section 2.1), this was reduced to a two-month window, August 1st to September 30th 2019, encompassing the M5.2 mainshock of September 4th 2019 and the wider unrest episode surrounding it. This reduction is stated plainly here, at the outset, rather than left for the reader to infer from the Methodology alone. It is the single largest limitation of this project, and is discussed as such in Section 4.7.

1.5 Terminology

A small number of technical terms recur throughout this dissertation and are defined here for clarity:

Magnitude of completeness (Mc): the minimum earthquake magnitude above which a catalogue is considered to record essentially all events that actually occurred, rather than missing an increasing proportion of smaller events (Wiemer and Wyss, 2000).

Gutenberg-Richter b-value: the slope of the linear relationship between earthquake magnitude and the logarithm of the cumulative number of earthquakes at or above that magnitude, describing how the relative frequency of large versus small earthquakes scales within a given catalogue or region.

Azimuthal gap: the largest angular gap, in degrees, between stations used to locate a given earthquake, as seen from the earthquake's hypocentre. A large azimuthal gap indicates poor station coverage around an event and is associated with less reliable location estimates.

Coalescence function: in QuakeMigrate specifically, the function used to measure how strongly waveform energy from across the network agrees on a common source location and origin time at each point in a search grid. A genuine earthquake produces a sharp peak in this function, both spatially and temporally.

Phase association: the process of grouping individual phase-arrival picks made at different stations, from methods such as EQTransformer or PhaseNet, into a self-consistent set of picks that plausibly originate from a single earthquake.

1.6 Study Region

Figure 1 shows the location of the TVZ study region within New Zealand, and New Zealand's own location globally, to orient readers unfamiliar with this setting before the detailed study area description that follows in the Methodology (Section 2.1).

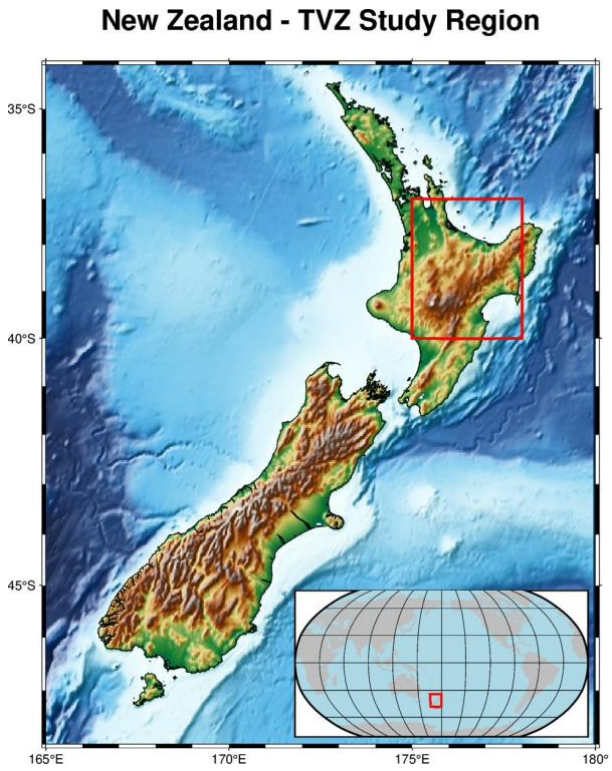


Figure 1: The TVZ study region (red box) within New Zealand, with an inset showing New Zealand's location globally.

2. Methodology

2.1 Study Area, Time Window and Data

The Taupō Volcanic Zone (TVZ) is a seismically and volcanically active rift in the central North Island of New Zealand, extending approximately 300km from Mount Ruapehu in the south to the offshore Bay of Plenty in the north (Seebeck et al., 2014). The zone experiences frequent small earthquakes linked to volcanic, geothermal and tectonic processes, but GeoNet's routine national catalogue only reliably detects events above a magnitude of approximately 2.0 in this region (Illsley-Kemp and Mestel, 2025), so most small earthquakes go undetected. This project reproduces and extends the automated detection workflow of Illsley-Kemp and Mestel (2025) using three independent tools, comparing their ability to build a more complete earthquake catalogue for the TVZ.

Continuous seismic waveform data were downloaded for 60 GeoNet stations within the TVZ using ObsPy (Beyreuther et al., 2010). Owing to computational issues with the High-performance Analytics and Learning Infrastructure (HALI) cluster, the study period was reduced from the proposed 17-year window (2007-2023) to a two-month period, August 1st to September 30th 2019. This period was chosen because it captures the 2019 Taupō volcanic unrest episode, including a magnitude 5.2 earthquake on September 4th 2019, providing a period of naturally elevated seismicity within a tractable computational timeframe. This is a substantial reduction in temporal scope from the original proposal, and its implications are considered throughout the results and discussion.

2.2 Comparative Framework

Three independent automated pipelines were built to detect, associate, locate and relocate earthquakes from the same waveform archive, allowing their outputs to be compared directly. Phase 1 combines the deep-learning phase picker EQTransformer (Mousavi et al., 2020) with the phase associator PyOcto (Münchmeyer, 2024), replicating the approach of Illsley-Kemp and Mestel (2025). Phase 2 combines the deep-learning phase picker PhaseNet (Zhu and Beroza, 2019) with the Bayesian associator GaMMA (Zhu et al., 2022), producing an independent catalogue built from different underlying algorithms. Both phases were then processed through an identical pipeline for absolute location, quality filtering, relative relocation and magnitude estimation, so that differences between the two final catalogues can be attributed to the detection and association stage rather than to inconsistencies in shared downstream methodology. A third, structurally different tool, QuakeMigrate, forms Phase 3 of the project and is described separately.

2.3 Phase Detection and Association

EQTransformer and PhaseNet are both convolutional neural networks trained to recognise the arrival of P and S seismic waves in a waveform, outputting a probability that a given sample corresponds to a genuine arrival. A pick is retained only where this probability exceeds a chosen threshold, so the threshold controls a trade-off between missing genuine small earthquakes, if set too high, and retaining false picks generated by noise, if set too low. Both models were deployed through SeisBench (Woollam et al., 2022), a common software framework that runs different pretrained picking models with a consistent interface. This is itself a methodological advantage for this comparison, since any differences in performance can be more confidently attributed to the underlying models rather than to inconsistencies in how they were run.

2.3.1 Phase 1: EQTransformer and PyOcto

EQTransformer (Mousavi et al., 2020) was run using its publicly available pretrained model, with parameters matching Illsley-Kemp and Mestel (2025): an overlapping 55-second analysis window; an event detection probability threshold of 0.2; and a phase pick probability threshold of 0.01. Duplicate picks arising from the overlapping windows were removed by keeping only the highest-probability pick within any 5-second window at a given station and phase. This produced 339,857 raw picks, of which 275,197 (81%) remained after this declustering step.

Declustered picks were associated into candidate earthquakes using PyOcto (Münchmeyer, 2024), which groups picks that are consistent with a common origin time and location. PyOcto requires a velocity model to estimate how long a seismic wave takes to travel between a candidate earthquake location and each station. The regional 1D velocity model used by Illsley-Kemp and Mestel (2025) is described in an unpublished doctoral thesis (Mestel, 2023) that could not be obtained. In its place, a single-layer velocity model was used, with a P-wave velocity of 5.7km/s and an S-wave velocity of 3.35km/s, based on independent seismic refraction studies of the TVZ crust (Stratford and Stern, 2010; Sherburn et al., 2003). This is a simplification relative to a fully resolved 1D or 3D velocity structure, and its implications are considered in the discussion. Association required a minimum of ten picks per candidate earthquake, at least five P and two S, yielding 2,982 candidate events from 63,957 associated picks (23% of the declustered total).

2.3.2 Phase 2: PhaseNet and GaMMA

PhaseNet (Zhu and Beroza, 2019) was deployed identically through SeisBench, because PhaseNet has no dedicated event-detection output, only P and S phase probabilities, a threshold sweep across five candidate values (0.05, 0.1, 0.2, 0.3 and 0.5) was carried out on a station subset to select an operating threshold. A threshold of 0.3 was chosen: this is SeisBench's own recommended default for this model, and it produced a per-station pick rate closest to that of EQTransformer, allowing a fairer comparison of the two catalogues' completeness. The full run produced 766,090 picks across all 60 stations, roughly 2.25 times more than EQTransformer at a comparable stage, consistent with PhaseNet's established characterisation in the literature as a higher-recall, lower-precision picker (Zhu and Beroza, 2019). Picks were declustered using the same 5-second rule as Phase 1, retaining 758,296 picks (99.0%).

Declustered PhaseNet picks were associated using GaMMA (Zhu et al., 2022), a Bayesian Gaussian mixture model associator that first groups picks using a density-based spatiotemporal clustering step before refining each cluster into a candidate earthquake. The

same homogeneous velocity values used for PyOcto were applied here for consistency between the two pipelines, and the same minimum-pick requirement was imposed. This produced 2,010 candidate events from 43,304 associated picks (5.7% of the declustered total).

2.4 Absolute Location

Candidate events from both pipelines were located using NonLinLoc (Lomax et al., 2000), which searches a three-dimensional grid for the hypocentre location and origin time that best fit the observed arrival times at each contributing station, using a probabilistic (Oct-tree) search algorithm and a time-weighted Equal Differential Time likelihood function.

NonLinLoc requires a velocity model to calculate expected travel times ahead of the location search. The original project design specified a fully three-dimensional velocity model built from the regional adjoint tomography of Chow et al. (2022). Attempting to build this model revealed a fundamental incompatibility, NonLinLoc's three-dimensional grid-building tool only accepts a small number of legacy tomographic inversion formats from the 1990s, and even selecting between them requires modifying and recompiling NonLinLoc's own source code. Given the scale of this undertaking relative to the time available, a one-dimensional depth profile was instead extracted directly from the same real Chow et al. (2022) tomography model, at a location representative of the study area, and used to build a horizontally uniform velocity grid. This is a simplification, the principal contribution of the Chow et al. (2022) model is its resolution of lateral velocity heterogeneity across the region, which this simplified profile necessarily discards. Its implications for location accuracy, particularly with depth, are considered in the discussion.

2.5 Quality Filtering

Following Illsley-Kemp and Mestel (2025), locations were retained only where the horizontal location error was less than 5km, the azimuthal gap between contributing stations was less than 180°, at least eight phase arrivals contributed to the solution, and the mean arrival-time residual was less than 0.5 seconds. Horizontal error was calculated from NonLinLoc's location covariance matrix as the square root of the sum of the horizontal variance terms, a standard practical approximation to the full confidence ellipse. Applying these filters retained 904 of 2,983 EQTransformer events (30.3%) and 933 of 2,011 PhaseNet events (46.4%). In both catalogues, azimuthal gap was the dominant rejection criterion, reflecting the geometry of the station network, which is concentrated around the region's volcanic centres rather than evenly distributed across the wider study area.

2.6 Relative Relocation

Quality-filtered events were relatively relocated using GrowClust3D.jl (Trugman et al., 2022), which improves the relative positions of nearby earthquakes using waveform cross-correlation. For each pair of nearby events, differential arrival times were measured directly from the seismic waveforms using EQcorrscan (Chamberlain et al., 2018), cross-correlating a short window around each pick after applying a 1-15 Hz bandpass filter, and retaining only correlations exceeding 0.65. This produced 2,313 usable event pairs for the EQTransformer catalogue and 3,649 for the PhaseNet catalogue.

As with the absolute location step, GrowClust3D.jl's ability to use a fully three-dimensional velocity structure depends on reading NonLinLoc's own travel-time grids, and its interpretation of these grids requires a map projection matching the one used to build them. NonLinLoc's grids in this project were built using its own "simple" local Cartesian projection, which is not among the small number of standard map projections GrowClust3D.jl's grid-reading code supports. Rather than reconstruct the location grids under a different, compatible projection, GrowClust3D.jl was instead run in its alternative mode, which computes travel times directly from the same simplified one-dimensional velocity profile used for absolute location, avoiding the incompatibility without requiring any new data. GrowClust3D.jl relatively relocated 423 of 904 EQTransformer events (46.8%) and 485 of 933 PhaseNet events (52.0%). The remaining events retain their absolute locations, having had too few reliable correlations to nearby events for relative relocation.

2.7 Local Magnitude Estimation

The original project design specified local magnitude (ML) estimation only for the QuakeMigrate catalogue, using an attenuation relationship from the same unpublished thesis (Mestel, 2023) unavailable for Section 2.3.1. Magnitude estimation was extended to all three catalogues, since a meaningful comparison of catalogue completeness depends on having consistent magnitude information for every event. In place of the unavailable relationship, a published, regionally calibrated local magnitude scale for the Taupō area was used (Illsley-Kemp et al., 2021), from the same research group responsible for the benchmark catalogue against which this project's results are compared. For each event, the instrument response was removed from the vertical-component waveform at each contributing station and the resulting trace was converted to the equivalent Wood-Anderson seismometer response (Anderson and Wood, 1925). The peak-to-trough amplitude of this simulated trace was then converted to a magnitude using:

$$ML = \log(A) - \log(A_0) + C$$

Where A is the Wood-Anderson amplitude in millimetres and $-\log(A_0)$ is a distance-dependent attenuation correction specific to the Taupō region (Illsley-Kemp et al., 2021). The station correction term, C, requires a calibration table not available from the published source, and was set to zero for all stations. This is a documented simplification rather than the full published method. The final magnitude for each event was taken as the median of all available per-station estimates.

2.8 Phase 3: QuakeMigrate

QuakeMigrate (Winder et al., 2026) takes a fundamentally different approach to the first two phases. Rather than picking phase arrivals at individual stations and then associating them into events, it migrates and stacks signal energy from every station across the whole network simultaneously, searching directly for times and locations where energy coherently aligns. This implicitly performs the equivalent of picking and association in a single step, and can succeed even where signals are too weak to pick reliably at any single station.

2.8.1 Travel-Time Lookup Table

QuakeMigrate requires its own travel-time lookup table to migrate waveform energy across the study grid. Unlike NonLinLoc, whose three-dimensional grid-building tool proved incompatible with the Chow et al. (2022) model (Section 2.4), QuakeMigrate's current released software only provides a ready-to-use utility for building this table from a one-dimensional velocity model, calling NonLinLoc's own travel-time solver internally. The same one-dimensional profile used throughout this project was reused directly for this step. This reflects QuakeMigrate's own documented, intended workflow rather than a further simplification. The lookup table was built at 2km node spacing across the same TVZ region used throughout.

2.8.2 Detection and Event Triggering

For each short time window, QuakeMigrate computes an onset function at every station using a short-term-average/long-term-average (STA/LTA) ratio, a standard technique that flags a likely arrival wherever a signal's short-term energy rises sharply above its recent background level. These onset functions are migrated across the lookup table and stacked to produce a single, continuous coalescence function describing how strongly, at each moment in time, energy from across the network coheres at some point in the grid. A coarser, four-times-decimated lookup table was used specifically for this computationally intensive step, since only the finer, full-resolution table is required for the refined location step that follows.

The coalescence function was searched for discrete candidate events using a normalised coalescence threshold, because this normalised value has no intuitive meaning ahead of time, the threshold was chosen empirically. Event counts were compared across a range of candidate values, and the threshold was selected as the value at which the resulting count first became consistent with the region's independently documented background seismicity rate of 50-200 events per month (Illsley-Kemp et al., 2022), rather than a value chosen for numerical convenience alone. Very low thresholds produced implausibly large event counts inconsistent with genuine seismicity, and values close to one particular level produced anomalous behaviour attributed to the coalescence trace itself hovering near that level. The threshold ultimately used avoided this region and produced 94 candidate events across the two-month study period, closely matching the low end of the documented background rate.

2.8.3 Location and Local Magnitude Estimation

Candidate events were relocated using QuakeMigrate's own refined sub-grid search, which fits a three-dimensional Gaussian function to the coalescence peak to obtain a location more precise than the underlying grid spacing, together with an independent covariance-based uncertainty estimate. Of 94 candidate events, 84 (89%) were successfully located.

Local magnitudes were estimated using the same Illsley-Kemp et al. (2021) Taupō-specific formula used for the EQTransformer and PhaseNet catalogues (Section 2.7), implemented as a custom function within QuakeMigrate's own magnitude module rather than its built-in generic option, keeping all three catalogues' magnitudes on a directly comparable scale. QuakeMigrate measures amplitude as half the peak-to-trough displacement, whereas the formula (and the other two catalogues) use the full peak-to-trough value. A constant correction of $\log_{10}(2)$ was added to the attenuation relationship to account for this, under the standard assumption that the two quantities differ by a factor of two for a broadly symmetric waveform.

2.8.4 Quality Filtering

The same four quality criteria used for the EQTransformer and PhaseNet catalogues (Section 2.5) were applied, with two thresholds recalibrated specifically for QuakeMigrate's own output rather than assumed to be numerically equivalent. QuakeMigrate reports two independent location uncertainty estimates: a narrower one from a Gaussian fit local to the coalescence peak; and a wider one from the spread of the full coalescence probability distribution above a percentile threshold. The horizontal error filter used the latter, since QuakeMigrate's own documentation identifies it as the quantity intended for quality filtering, and it is conceptually equivalent to the covariance-based error used for the other two catalogues. QuakeMigrate's residual is also computed differently to NonLinLoc's own optimisation-based residual,

comparing predicted and independently re-picked arrival times across the whole network rather than the residual of a single best-fitting solution, and was found to sit on a substantially different absolute scale. The residual threshold was therefore recalibrated to 1.0 seconds, close to the median of the observed distribution, preserving the same qualitative role of rejecting the more poorly-constrained half of events rather than assuming numerical equivalence with the fixed 0.5 second threshold used elsewhere. Applying all four criteria retained 31 of 84 located events (36.9%).

2.9 Comparative Analysis Methods

Following the construction of all three catalogues (Sections 2.3-2.8), a common set of comparative analysis methods was applied to enable direct benchmarking between them and against an independent reference catalogue.

Each quality-filtered catalogue was compared against the benchmark catalogue of Illsley-Kemp and Mestel (2025), matching events within a 2-second origin-time window and 5km horizontal separation. A matched event pair was counted as a true positive, an unmatched catalogue event as a false positive, and a benchmark event with no matching catalogue event as a false negative, from which precision, recall and an event-count ratio were calculated for each catalogue.

Magnitude of completeness (Mc) was estimated using two independent, established methods (Wiemer and Wyss, 2000): maximum curvature (MAXC), which identifies Mc as the magnitude bin containing the greatest number of events in the non-cumulative frequency-magnitude distribution; and a Goodness-of-Fit test (GFT), which compares the observed cumulative frequency-magnitude distribution above successively higher magnitude cutoffs against the synthetic distribution predicted by a fitted Gutenberg-Richter relationship, taking Mc as the lowest cutoff at which a chosen confidence level is reached. The standard 90% confidence threshold proved too strict for these sparser regional catalogues and was relaxed to 80%, a deviation applied consistently across all catalogues rather than selectively. Gutenberg-Richter b-values were estimated above each catalogue's own MAXC-derived Mc using maximum likelihood estimation as the point estimate, with uncertainty quantified through 1,000 bootstrap resamples. Bootstrap resampling was used only for uncertainty quantification and not as the point estimate itself, since averaging across resampled b-values would introduce bias for this nonlinear statistic.

Sub-regional classification (Section 3.6) assigned each event to the nearest of four 20km radius circles centred on documented volcanic centres (Taupō, Ōkataina, Tongariro, Ruapehu), or to an "Other" category if outside all four. Magnitude-frequency distributions were

plotted only for sub-region/catalogue combinations containing at least 5 events, following standard practice for avoiding statistically unreliable distributions built from too few observations.

Kernel density estimation (KDE) was used to convert each catalogue's discrete event locations into a continuous spatial seismicity rate surface (Sections 3.9-3.11). A direct, manually implemented two-dimensional isotropic Gaussian kernel was used, rather than a general-purpose statistical package's own KDE implementation, because such packages typically set kernel bandwidth relative to the data's own spread rather than accepting an absolute physical value. Here, bandwidth was set explicitly to each catalogue's own median horizontal location uncertainty, so that each catalogue's density surface reflects its own genuine location precision. This implementation was verified against synthetic data with known true properties before use on real data, confirming correct normalisation and correct behaviour of the density estimate at a location of known true density. Resulting rate surfaces (in events per km² per year) were visualised on a logarithmic colour scale, since a linear scale calibrated to the sharp, narrow peak typical of these catalogues compressed the great majority of genuine spatial variation into visual indistinguishability. Difference maps between catalogue pairs were computed as $\log_{10}$(rate A / rate B) on a shared spatial grid, a symmetric quantity directly interpretable as the number of orders of magnitude by which one catalogue's inferred rate exceeds the other's at a given location.

Depth cross-sections (Section 3.13) were constructed along two transects: an along-TVZ transect from Ruapehu to Ōkātaina, using the true geodetic azimuth between these two sub-region centres and extended by 20km at each end; and a cross-TVZ transect perpendicular to this, centred on Taupō. Each event's position along a given transect, and its perpendicular distance from it, was computed from its true geodetic distance and bearing from the transect's start point. Only events within 20km of a transect's line were included in that transect's profile.

3. Results

3.1 Catalogue Overview

Candidate event counts, quality-filter retention and relative relocation rates varied substantially between the three pipelines (Table 1). Phase 1 (EQTransformer/PyOcto) produced 2,982 candidate events, of which 904 (30.3%) passed quality filtering, and 423 of these (46.8%) were subsequently relatively relocated. Phase 2 (PhaseNet/GaMMA) produced 2,010 candidate events, retaining 933 (46.4%) after quality filtering and relatively relocating 485 (52.0%). Phase

3 (QuakeMigrate) produced 94 candidate events, considerably fewer than either other pipeline, of these, 84 (89.4%) were successfully located, and 31 of the located events (36.9%) passed quality filtering.

Table 1: Candidate, located and quality-filtered event counts by pipeline.

	EQTransformer	PhaseNet	QuakeMigrate
Candidate events	2,982	2,010	94
Located events	n/a	n/a	84 (89.4%)
Quality-filtered events	904 (30.3%)	933 (46.4%)	31 (36.9%)
Relatively relocated events	423 (46.8%)	485 (52.0%)	n/a

3.2 Benchmark Comparison

All three quality-filtered catalogues were compared against the Illsley-Kemp and Mestel (2025) benchmark catalogue (1,030 events within the study window), matching events within ±2 seconds and 5km (Table 2). EQTransformer achieved a precision of 61.2% and a recall of 53.7%. PhaseNet achieved a precision of 57.1% and a recall of 51.6%. QuakeMigrate achieved a precision of 83.9% and a recall of 2.5%.

Table 2: Benchmark comparison against Illsley-Kemp and Mestel (2025).

	EQTransformer	PhaseNet	QuakeMigrate
Precision	61.2%	57.1%	83.9%
Recall	53.7%	51.6%	2.5%
Event count ratio (vs. benchmark)	0.88	0.91	0.03

3.3 Magnitude of Completeness and b-value

Magnitude of completeness (Mc) and Gutenberg-Richter b-values were estimated for all three catalogues using two independent methods: maximum curvature (MAXC); and a Goodness-of-Fit test, with b-value uncertainty estimated through 1,000 bootstrap resamples (Table 3). MAXC gave Mc values of 1.55 (EQTransformer), 1.45 (PhaseNet) and 1.95 (QuakeMigrate). The Goodness-of-Fit method gave a lower Mc in every catalogue: 1.40 (EQTransformer); 1.10 (PhaseNet); and 1.70 (QuakeMigrate). Using the MAXC Mc, bootstrap b-values were 1.474 ± 0.073 (EQTransformer, n=380 events above Mc), 1.594 ± 0.103 (PhaseNet, n=290) and 0.729 ± 0.177 (QuakeMigrate, n=31). The benchmark catalogue's own Mc and b-value, computed identically over the same window, were 0.85 and 0.908 respectively.

Table 3: Magnitude of completeness and Gutenberg-Richter b-value by catalogue.

	EQTransformer	PhaseNet	QuakeMigrate	Benchmark
Mc (MAXC)	1.55	1.45	1.95	0.85
Mc (Goodness-of-Fit)	1.40	1.10	1.70	n/a
b-value ± bootstrap SD	1.474 ± 0.073	1.594 ± 0.103	0.729 ± 0.177	0.908
n events above Mc (MAXC)	380	290	31	n/a

3.4 Spatial and Depth Distribution

Mean hypocentral depths were broadly similar across catalogues (Table 4): 14.80 ± 9.63km (EQTransformer); 12.28 ± 8.17km (PhaseNet); and 12.18 ± 5.06km (QuakeMigrate). In all three catalogues, events not matched to the benchmark catalogue had a systematically greater mean depth than the catalogue as a whole: 19.24km against an overall mean of 14.80km for EQTransformer; 14.01km against 12.28km for PhaseNet; and 17.75km against 12.18km for QuakeMigrate. Azimuthal gap was the dominant quality-filter rejection criterion in all three catalogues.

Table 4: Depth statistics and unmatched-event depth comparison by catalogue.

	EQTransformer	PhaseNet	QuakeMigrate
Mean depth, all events (km)	14.80 ± 9.63	12.28 ± 8.17	12.18 ± 5.06
Mean depth, unmatched events (km)	19.24	14.01	17.75
Mean horizontal error (km)	1.48	1.41	3.83

3.5 Cross-Catalogue Agreement

Of EQTransformer's 904 quality-filtered events, 621 (68.7%) were also present in the PhaseNet catalogue. All 31 of QuakeMigrate's quality-filtered events were independently found in both the EQTransformer and PhaseNet catalogues (100%).

3.6 Sub-Region Distribution and Magnitude-Frequency

Events were classified into four sub-regions (Taupō, Ōkataina, Tongariro and Ruapehu: 20km radius circles centred on documented coordinates) or “Other” if outside all four (Table 5). The Taupō sub-region contained the largest share of events in every catalogue. QuakeMigrate's quality-filtered events were concentrated almost entirely within the Taupō sub-region (29 of 31 events, 93.5%), with only one event each falling within the Ōkataina sub-region and the “Other” category, and none within Tongariro or Ruapehu.

Table 5: Event counts by sub-region and catalogue.

Sub-region	EQTransformer	PhaseNet	QuakeMigrate
Taupō	398	420	29
Ōkataina	52	78	1
Tongariro	51	57	0
Ruapehu	58	87	0
Other	345	291	1

Magnitude-frequency distributions were plotted for each catalogue within each sub-region containing at least five events (Figure 2). Three sub-region/catalogue combinations were excluded on this basis, all in the QuakeMigrate catalogue: Ōkataina (1 event); Tongariro (0 events); and Ruapehu (0 events).

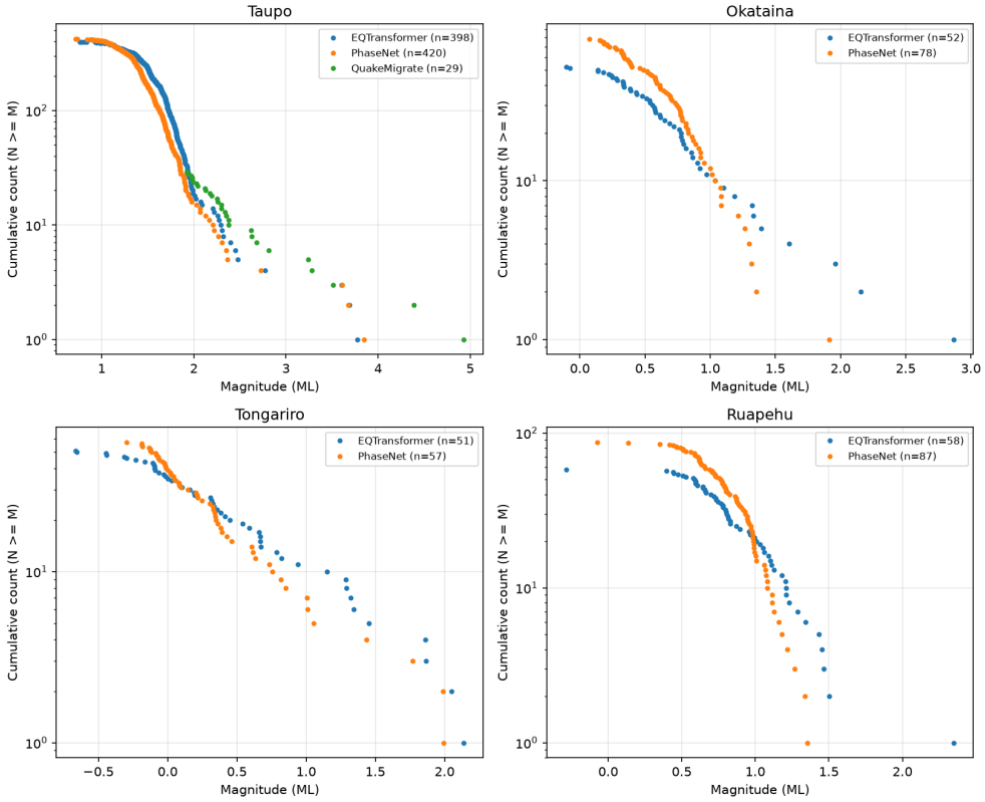


Figure 2: Magnitude-frequency distributions by sub-region and catalogue. Sub-region/catalogue combinations with fewer than 5 events are omitted.

3.7 Temporal Distribution

The 94 candidate events were not evenly distributed across the two-month study period (Table 6). August contained 28 events (0.90 events per day), September contained 66 events (2.20 events per day). This overall difference was concentrated almost entirely within a 3-day window, September 4th-6th, which alone contained 37 events (12.33 events per day), including

25 events on September 4th, the date of the M5.2 mainshock. Excluding this 3-day window, the remaining September activity (29 events over 27 days, 1.07 events per day) was closely comparable to the August rate (0.90 events per day), a 1.19-fold difference.

Table 6: Temporal distribution of the 94 candidate events.

Period	Events	Days	Rate (events/day)
August (full month)	28	31	0.90
September 4th-6th	37	3	12.33
September, excl. Sept 4th-6th	29	27	1.07

3.8 Seismicity Maps

Geographic seismicity maps were produced for each catalogue (Figures 3-5), showing individual event locations coloured by depth and sized by magnitude, overlaid on real topography and mapped fault traces from the New Zealand Active Faults Database (Langridge et al., 2016). Fault density was visibly highest around Ōkātaina and along the central rift corridor, consistent with the known structural complexity of this part of the TVZ. In both the EQTransformer and PhaseNet maps, a dense, tightly-clustered concentration of events was visible immediately around Taupō, consistent with the September 4th mainshock sequence identified in Section 3.7. The QuakeMigrate map showed a comparably sparse, spatially concentrated distribution, with the great majority of its 31 events clustered close to Taupō.

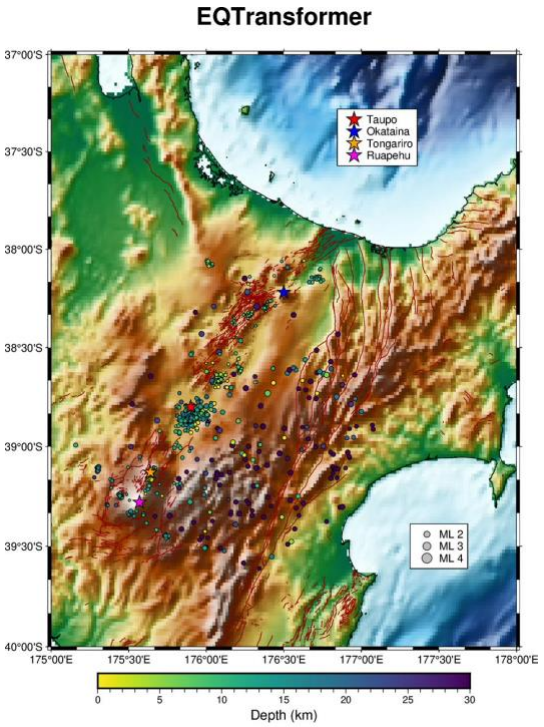


Figure 3: EQTransformer seismicity map, with real topography, NZAFD fault traces, and sub-region markers.

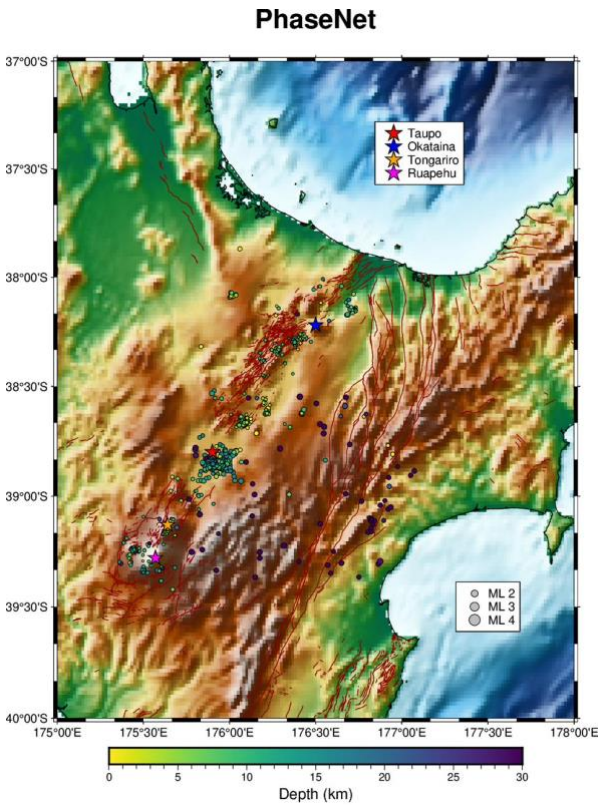


Figure 4: PhaseNet seismicity map, with real topography, NZAFD fault traces, and sub-region markers.

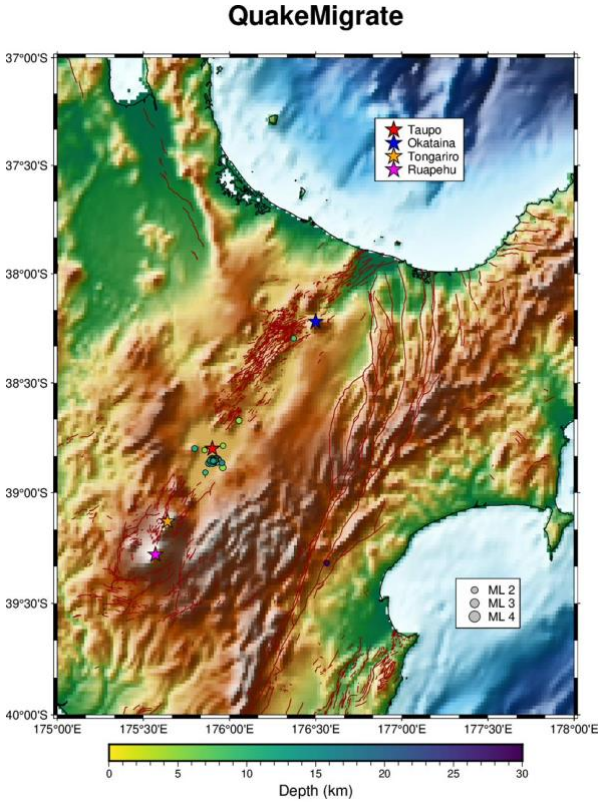


Figure 5: QuakeMigrate seismicity map, with real topography, NZAFD fault traces, and sub-region markers.

3.9 Seismic Density Mapping

Kernel density estimation (KDE) was used to convert each catalogue's discrete event locations into a continuous seismicity rate surface (events per km² per year), using a Gaussian kernel with bandwidth set to each catalogue's own median horizontal location uncertainty (Figures 6-8). This gave bandwidths of 1.18km (EQTransformer), 1.17km (PhaseNet) and 3.84km (QuakeMigrate). In all three catalogues, the rate surface reached its peak value directly over Taupō. Peak rates were substantially higher in the EQTransformer and PhaseNet surfaces than in the QuakeMigrate surface, consistent with their far larger event counts.

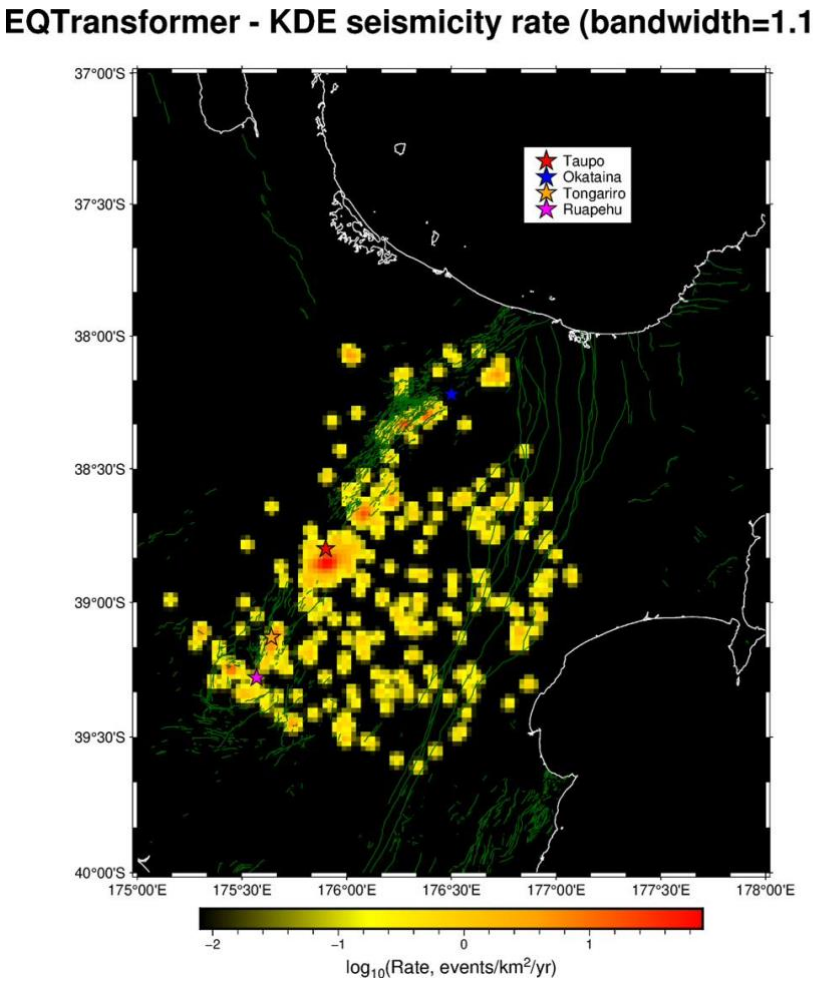


Figure 6: EQTransformer KDE seismicity rate surface (log_{10} scale, bandwidth 1.18km).

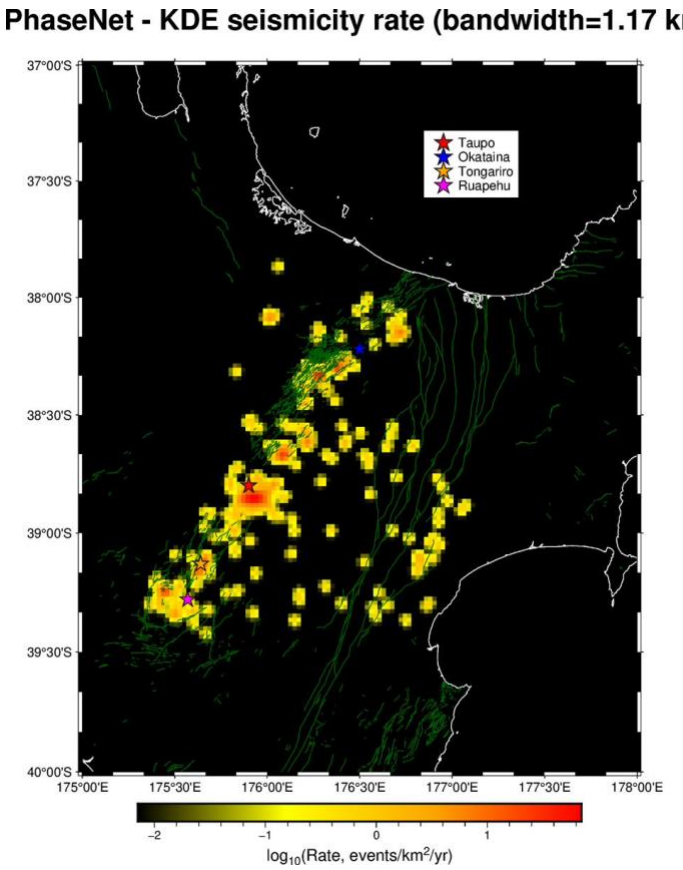


Figure 7: PhaseNet KDE seismicity rate surface (log_{10} scale, bandwidth 1.17km).

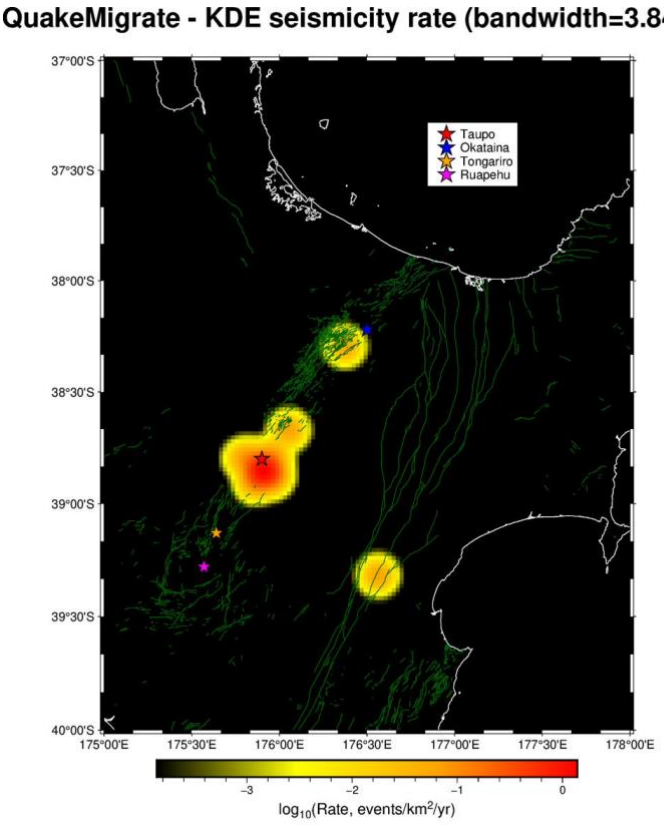


Figure 8: QuakeMigrate KDE seismicity rate surface (log_{10} scale, bandwidth 3.84km).

3.10 Depth-Stratified and Magnitude-Filtered Density

Separate KDE density surfaces were computed for shallow (0-15km) and mid-crustal (15-40km) depth bins, and for magnitude bins spanning each catalogue's own Mc to M2.5 and M2.5 and above. Event counts and bandwidths for all subsets are given in Table 7. Depth-stratified surfaces for the EQTransformer and PhaseNet catalogues (Figures 9-10) retained a broadly similar spatial footprint at both depth ranges. The equivalent QuakeMigrate surfaces are not shown, as its mid-crustal bin contained only three events. The M2.5+ bin was sparse in every catalogue (6 events, EQTransformer; 4, PhaseNet; 9, QuakeMigrate), an expected consequence of the Gutenberg-Richter relationship rather than a data quality issue. The magnitude-filtered surfaces are likewise not shown for this reason, though in both the EQTransformer and PhaseNet catalogues the M2.5+ population was more tightly concentrated around Taupō than the Mc-to-M2.5 population, which extended more broadly across all four sub-regions.

Table 7: Event counts and KDE bandwidths for depth and magnitude-stratified subsets.

Subset	EQTransformer	PhaseNet	QuakeMigrate
0-15km depth	494 (1.10km)	681 (1.08km)	28 (3.81km)
15-40km depth	410 (1.29km)	252 (1.44km)	3 (4.34km)
Mc-M2.5	324 (1.24km)	239 (1.24km)	20 (4.18km)
M2.5+	6 (1.03km)	4 (1.19km)	9 (3.29km)

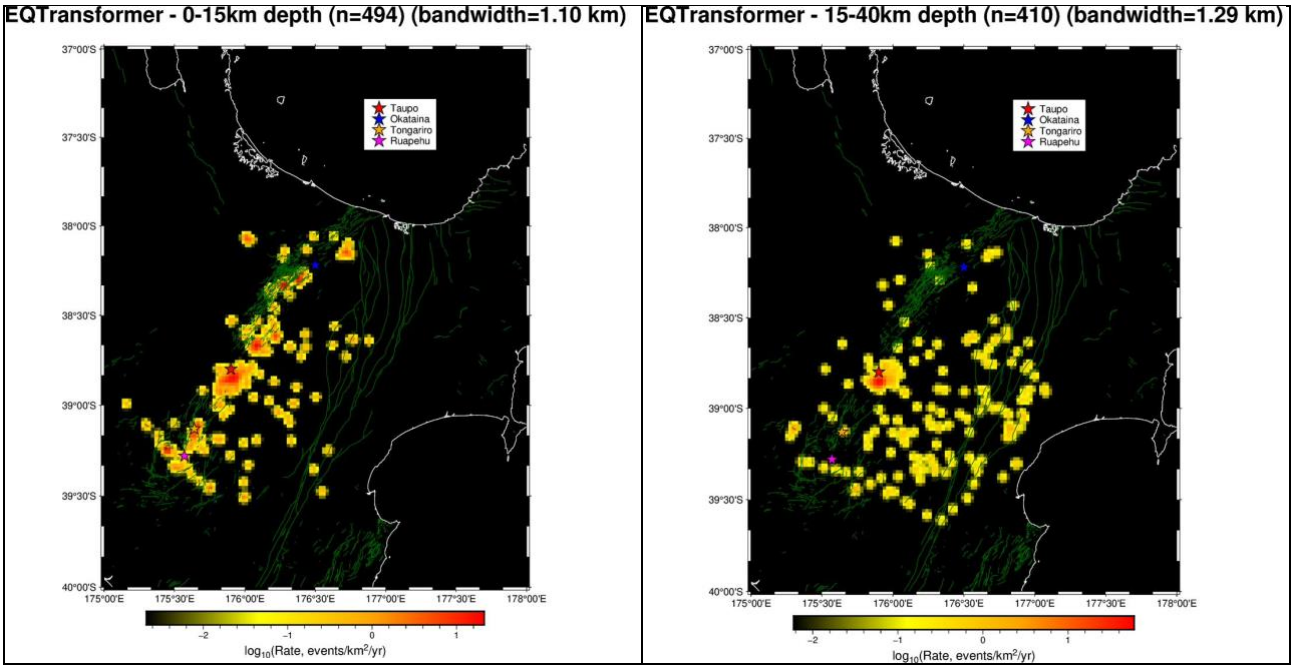


Figure 9: EQTransformer KDE density, 0-15km (left) and 15-40km (right) depth.

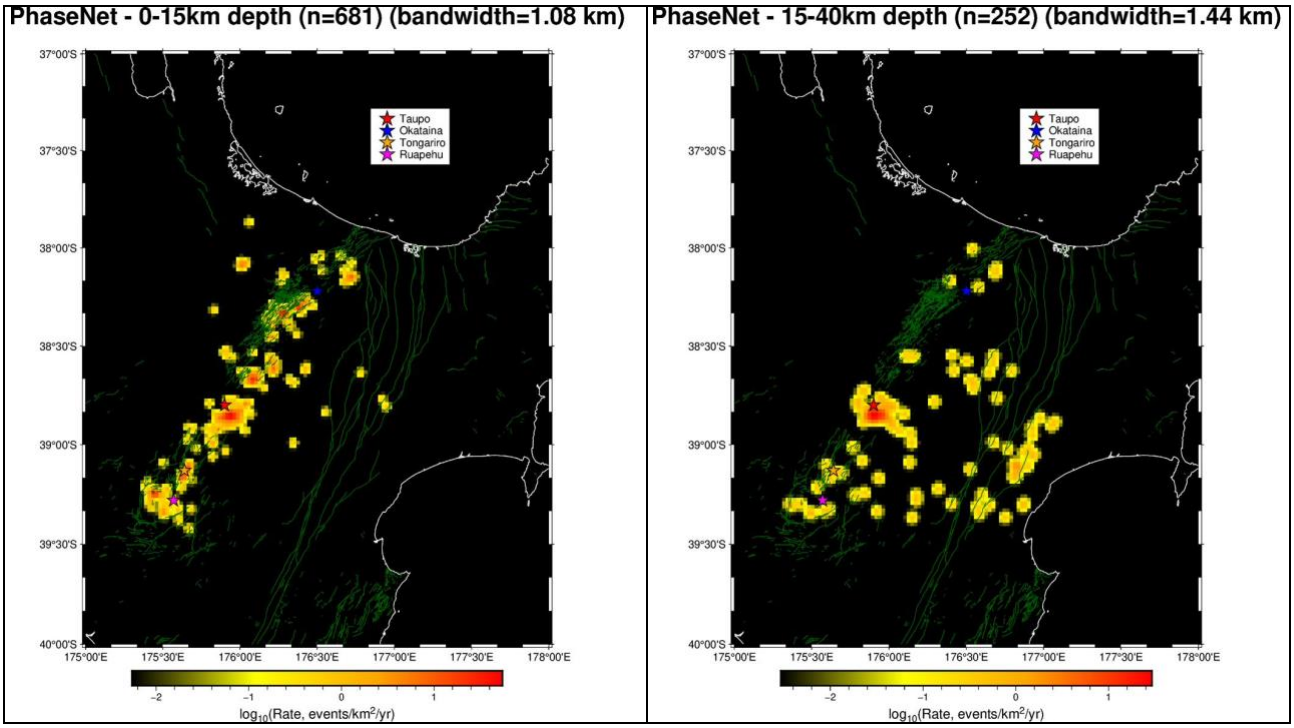


Figure 10: PhaseNet KDE density, 0-15km (left) and 15-40km (right) depth.

3.11 Catalogue Difference Maps

Difference maps were computed for each catalogue pair as $\log_{10}$(rate A / rate B) on a shared spatial grid, a symmetric quantity in which positive values indicate a higher relative rate in the first-named catalogue and negative values a higher rate in the second. The EQTransformer-PhaseNet difference (Figure 11) showed a mixed pattern of localised positive and negative patches across the study region, without either catalogue dominating overall. Both catalogue pairs involving QuakeMigrate (not shown) showed a predominantly positive pattern across most of the region, with QuakeMigrate-dominant patches confined to the immediate vicinity of its own event clusters. This difference is driven primarily by the roughly thirty-fold disparity in retained event count rather than by genuine spatial disagreement between the methods.

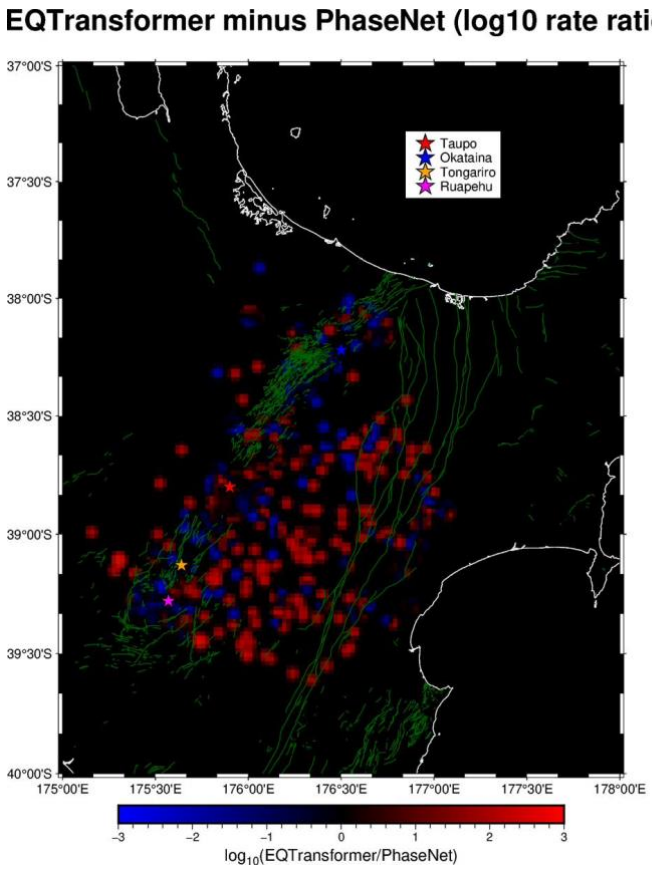


Figure 11: Difference map, EQTransformer minus PhaseNet (log_{10} rate ratio).

3.12 Extended Temporal Analysis

Daily event counts were compared across all three catalogues over the full study period (Figure 12), and the mainshock-window analysis of Section 3.7 was repeated for EQTransformer and PhaseNet (Table 8). Both catalogues showed the same qualitative pattern as QuakeMigrate: a pronounced spike in the September 4th-6th window (89.0 and 90.3 events per day respectively, against August rates of 12.0 and 12.9 events per day). Unlike QuakeMigrate, however, both EQTransformer and PhaseNet showed a reduction in September activity once the mainshock window was excluded, relative to August (ratios of 0.80 and 0.75 respectively), whereas QuakeMigrate's equivalent ratio was 0.98, indicating an almost identical rate in both periods.



Figure 12: Daily event counts, all three catalogues, with the September 4th mainshock marked.

Table 8: Mainshock-window comparison across all three catalogues.

	EQTransformer	PhaseNet	QuakeMigrate
August rate (events/day)	12.03	12.90	0.23
Sept. 4th-6th rate (events/day)	89.00	90.33	6.00
Sept. (excl. 4th-6th) rate (events/day)	9.67	9.63	0.22
Ratio: Sept. (excl.) / August	0.80	0.75	0.98

3.13 Depth Cross-Sections

Two depth cross-sections were constructed: an along-TVZ transect from Ruapehu to Ōkātaina (extended 20km at each end, total length 182.8km), and a cross-TVZ transect perpendicular to this, centred on Taupō (total length 80km). Events within 20km of each transect line were included (Figures 13-14). The along-TVZ cross-section showed several spatially distinct clusters along the transect, corresponding to the Ruapehu, Tongariro, Taupō and Ōkātaina sub-regions, separated by intervals of markedly lower event density. The cross-TVZ cross-section showed a single, well-constrained cluster centred on the transect, with the great majority of events between approximately 5 and 20km depth.

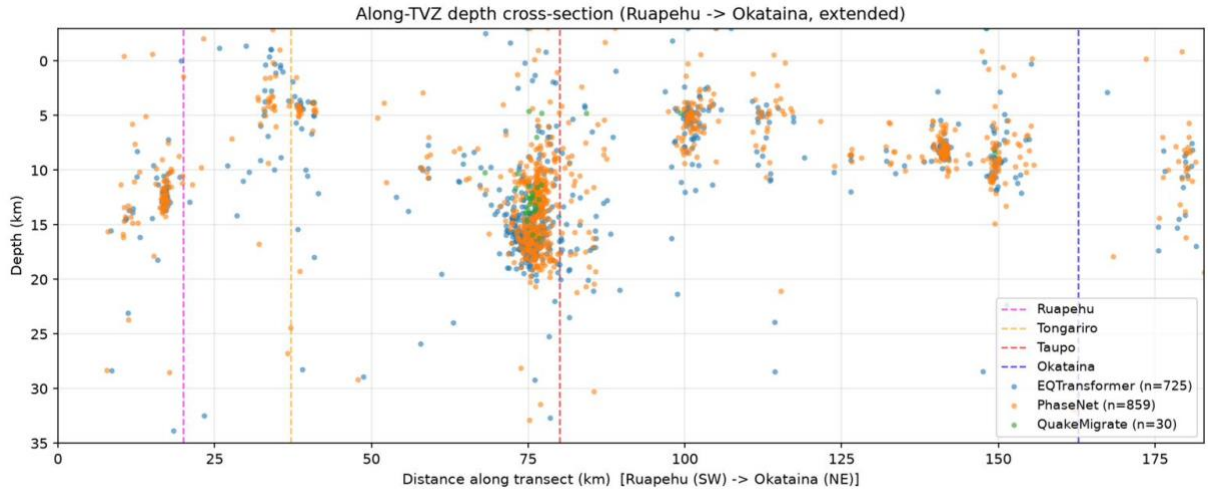


Figure 13: Along-TVZ depth cross-section (Ruapehu to Ōkātaina, extended).

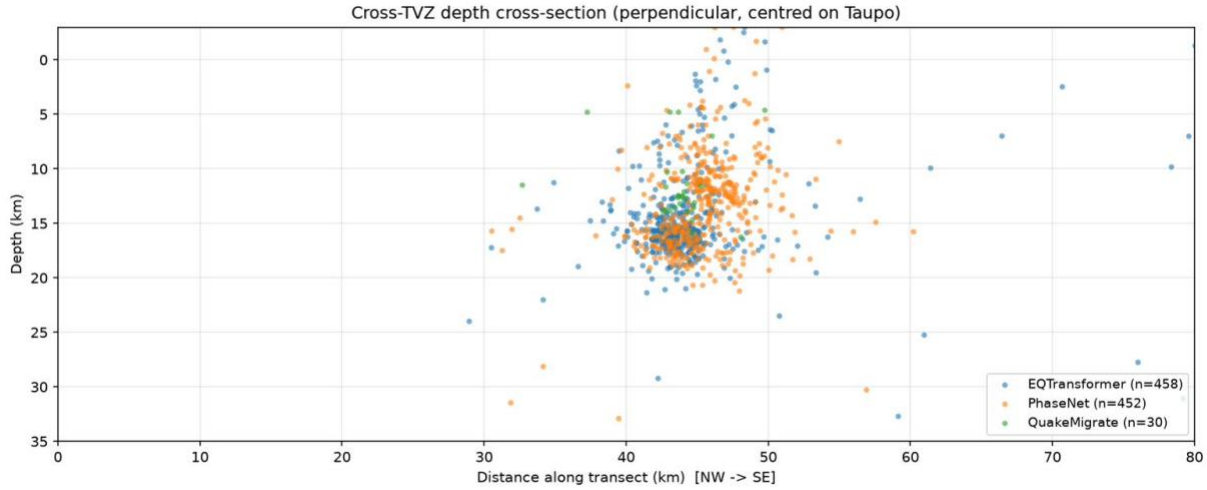


Figure 14: Cross-TVZ depth cross-section (perpendicular, centred on Taupō).

4. Discussion

4.1 Reproducibility of Illsley-Kemp and Mestel (2025)

RQ1 asked whether the workflow of Illsley-Kemp and Mestel (2025) could be reproduced. The honest answer is a partial reproduction. EQTransformer, run with the same picker, association minimum-pick requirements and quality-filtering criteria as the original study, recovered 88% of the benchmark catalogue's own event count within the study window and matched 53.7% of individual benchmark events directly, with 61.2% of its own retained events matching a genuine benchmark event (Table 2). This is a substantial, genuine overlap, but not a full reproduction, and the gap is attributable to specific, documented methodological deviations.

Three deviations account for most of this gap. First, the original study's regional velocity model is described only in an unpublished doctoral thesis (Mestel, 2023) that could not be obtained. A simplified, single-layer velocity model derived from independent seismic refraction studies of the TVZ crust was used in its place throughout this project (Section 2.3.1). Second, PyOcto's phase association in this project used a homogeneous velocity assumption for computational tractability, rather than the fully resolved three-dimensional structure the original study had available. Third, GrowClust3D.jl's relative relocation was run in its one-dimensional travel-time mode (Section 2.6), because its three-dimensional mode depends on reading NonLinLoc's own travel-time grids under a map projection GrowClust3D.jl's grid-reading code does not support. Each of these is a substantive simplification relative to the original study's own three-dimensional approach, and each would be expected to reduce location precision and therefore reduce the proportion of events correctly matched to the benchmark within the tight 5km, 2-second matching window used here.

This gap is itself a substantive finding of this project, not simply a limitation. It demonstrates concretely how much a benchmark comparison of this kind depends on velocity-model fidelity specifically, separate from the underlying phase-picking and association algorithms themselves. PhaseNet, run through an entirely independent picker and associator (GaMMA) but subject to the same velocity-model simplifications, achieved a closely comparable precision (57.1%) and recall (51.6%) to EQTransformer, consistent with the velocity model, rather than the choice of picker, being a major shared driver of the gap between this project's reproduction and the original study's own reported performance.

4.2 Convergent Methodological Findings

The single most consistent finding across this project is a systematic depth-degradation pattern, found independently in all three catalogues despite their structurally different detection philosophies. In every catalogue, events that failed to match the benchmark catalogue had a substantially greater mean depth than the catalogue as a whole: 19.24km against an overall mean of 14.80km for EQTransformer; 14.01km against 12.28km for PhaseNet; and 17.75km against 12.18km for QuakeMigrate (Table 4). This pattern is now supported by a third, independent line of evidence, the depth cross-sections (Section 3.13) show a well-constrained, tightly clustered depth distribution in the cross-TVZ profile, consistent with the shallower, better-located population dominating the located catalogues, while the along-TVZ profile's distinct, spatially separated clusters are each internally coherent in depth rather than diffuse.

EQTransformer, PhaseNet and QuakeMigrate share almost nothing in their underlying detection approach, but they do share two things: the same simplified one-dimensional velocity

model (Section 2.4); and the same real station network geometry. That three structurally independent tools converge on the same depth-degradation signature is stronger evidence that this reflects a genuine limitation of the shared velocity model and network geometry, rather than a quirk of any single tool's own algorithm. A one-dimensional velocity model necessarily cannot capture real lateral heterogeneity in seismic velocity with depth, and travel-time errors from this simplification would be expected to bias depth estimates specifically, more than horizontal position, for exactly the deeper, more poorly constrained events that are also least likely to be independently confirmed by a benchmark catalogue built with better velocity information.

A second convergent finding is that azimuthal gap was the dominant quality-filter rejection criterion in all three catalogues (Sections 2.5, 2.8.4, 3.4). This is consistent with the real geometry of the GeoNet station network in the TVZ, which is concentrated around the region's volcanic centres rather than evenly distributed across the wider study area (Section 2.5), and is now also directly visible in the fault-overlaid seismicity maps produced in this project (Section 3.8), where station coverage is visibly denser near Taupō and Ōkātaina than in the surrounding rift corridor. This is a physical property of the monitoring network rather than a processing artefact, and would be expected to affect any automated detection method applied to this same network in the same way.

4.3 Precision/Recall Trade-offs Across Tools

EQTransformer and PhaseNet's own established characterisations in the literature, as a moderate-recall/moderate-precision picker and a comparatively higher-recall picker respectively (Zhu and Beroza, 2019), showed up empirically at the raw picking stage. PhaseNet produced roughly 2.25 times more raw picks than EQTransformer from the same waveform archive (766,090 against 339,857: Section 2.3.2). This ordering did not, however, persist through to final recall against the benchmark catalogue, where EQTransformer's recall (53.7%) was marginally higher than PhaseNet's (51.6%: Table 2). This is an informative finding, it demonstrates that a picker's raw output rate and its final, benchmarked recall are not the same thing, since both catalogues pass through identical downstream association, location and quality-filtering stages that can materially change their relative ordering. A picker generating more raw picks does not automatically translate into more genuine, well-located events surviving to the final catalogue if a larger share of those additional picks are lower-quality.

QuakeMigrate's own precision/recall profile is a far more extreme trade-off than either picker-based catalogue: 83.9% precision against just 2.5% recall (Table 2). This is a direct, expected

consequence of the conservative detection threshold chosen for this project (Section 2.8.2), selected specifically to keep the resulting event count consistent with the TVZ's independently documented background seismicity rate rather than for numerical convenience. A lower threshold would recover more genuine events, at the cost of admitting substantially more noise-driven false detections, as demonstrated directly by the noise-resonance anomaly encountered during threshold selection (Section 4.4). This trade-off has a measurable cost, made visible directly by this project's own extended temporal analysis (Section 3.12). EQTransformer and PhaseNet both independently detected a roughly 20-25% reduction in background activity during September once the September 4th-6th mainshock window is excluded, relative to August (ratios of 0.80 and 0.75: Table 8), while QuakeMigrate's own equivalent ratio (0.98) shows almost no such decline. This is not because the decline did not occur, but because QuakeMigrate's conservative threshold is simply not sensitive enough to detect it: it reliably captures the largest, most energetic events regardless of when they occur, but is correspondingly blind to genuine, smaller-magnitude changes in background activity that the two picker-based catalogues, with their much higher volume of retained events, are able to resolve.

4.4 The QuakeMigrate Threshold-Selection Story

QuakeMigrate's own threshold-selection process, described in Section 2.8.2, surfaced two methodological lessons. First, an initial threshold sensitivity sweep was found, on close inspection, to be contaminated. QuakeMigrate only overwrites a given day's output file when new events are found at a new threshold, so earlier, more permissive threshold runs' output silently persisted into later, stricter threshold runs that should have produced different results. This was only caught by comparing file modification timestamps against trigger summary reports and finding days reporting zero newly triggered events despite an output file containing thousands of rows from an earlier run. The entire threshold sweep was invalid until each run began from a clean output directory. This is a straightforward, avoidable software behaviour, but one with a silent capacity to produce a plausible-looking but entirely wrong result if not specifically checked for.

Second, a threshold value close to a specific coalescence level produced anomalously large, implausible event counts, attributed to the underlying coalescence trace itself hovering near that level and triggering on essentially noise-driven crossings rather than genuine seismic energy. A subsequent, independent investigation, prompted by a reasonable concern that the final retained catalogue (31 events) might be too small for robust statistical analysis, tested a lower, more permissive threshold (1.3 rather than 1.4) directly. This produced a larger raw event count, but quality-filtering revealed the additional, marginal population was

overwhelmingly low-quality (9.4% retention against 36.9% at the threshold ultimately used, with roughly eight-fold worse mean horizontal error), confirming that the noise-resonance anomaly reflected a limitation of static coalescence thresholding rather than unwarranted caution. Both findings point to the same underlying lesson, static, manually chosen detection thresholds are a limitation of this kind of network-migration approach in a noisy, volcanically and geothermally active setting, and a dynamic thresholding approach, for example one based on the median absolute deviation of the coalescence trace itself rather than a single fixed value, is a candidate for future work (Section 5).

4.5 The August/September Activity Pattern

The clearest temporal signal in this project's results is the concentration of activity around the September 4th 2019 M5.2 mainshock. All three catalogues independently show a pronounced rate spike in the September 4th-6th window, and EQTransformer and PhaseNet additionally show a comparable decline in background activity through the remainder of September relative to August (Section 3.12). Two explanations are worth weighing against one another: that this reflects a genuine, physical unrest signal in TVZ seismicity; or that it is an artefact of detection sensitivity or data availability changing over the study window.

The genuine-unrest interpretation is well supported by independent, published evidence specific to this exact period. Illsley-Kemp et al. (2021) characterise 2019 as a documented period of volcanic unrest at Taupō specifically, identified by multiple seismic swarms and ground deformation. More specifically still, Aber et al. (2025) document 10 discrete spatiotemporal earthquake swarms along the northern and central TVZ spanning March to September 2019, a period that entirely encompasses this project's own two-month study window. This is an important qualification for this project's own August/September comparison, August was not a quiet, pre-unrest baseline period, but was itself already part of this same, ongoing, independently documented unrest sequence, several months after its own onset. The comparison made in this project (Section 3.12) is therefore better understood as elevated mainshock-adjacent activity against an already-elevated background, rather than mainshock-driven activity against a genuinely quiet control period, and this project's own results should be read with that framing.

An artefact-driven explanation cannot be fully ruled out on this project's own evidence alone, since station availability or ambient noise conditions were not independently audited across the study window as part of this project. However, the fact that three structurally independent detection methods, and multiple independent, published sources describing this same period using entirely different data and methods, all converge on the same qualitative pattern is

considerably stronger evidence for a genuine physical signal than any single one of these sources alone would provide.

4.6 Implications for Hazard Characterisation and Catastrophe Modelling

This project's results connect directly to currently-operational hazard characterisation infrastructure, not only to the academic literature. Moody's RMS states that it is a member of the technical advisory committee for the 2022 Aotearoa New Zealand National Seismic Hazard Model (“GNS22”), and that its New Zealand Earthquake HD Model Version 3 was updated specifically to incorporate this revision (Moody's RMS, 2024). The Seismicity Rate Model underlying GNS22 (Gerstenberger et al., 2022) characterises background seismicity, away from mapped faults, using a hybrid model that combines earthquake-catalogue-based smoothed seismicity, geodetic strain rate, and fault slip-rate data; a separate uniform rate zone component is used specifically for the lowest-seismicity regions, where the earthquake catalogue alone is considered too incomplete to constrain rates reliably. The results of this project, three independent, quality-controlled TVZ earthquake catalogues with lower magnitudes of completeness than GeoNet's routine catalogue, are precisely the kind of input the catalogue-based component of an operational national hazard model depends on.

The magnitude of completeness achieved varied by catalogue and estimation method, but was consistently and substantially lower than GeoNet's routine detection threshold of approximately Mc 2.0 in this region (Illsley-Kemp and Mestel, 2025): MAXC gave Mc 1.55 (EQTransformer); 1.45 (PhaseNet); and 1.95 (QuakeMigrate); the Goodness-of-Fit method gave lower values still, 1.40, 1.10 and 1.70 respectively. This matters substantially, not just in principle. The GNS22 Seismicity Rate Model documentation explicitly identifies a low-detection bias in the New Zealand earthquake catalogue in lower-rate regions as a known limitation, one it currently addresses using geodetic strain rate as a proxy where catalogue completeness is judged insufficient. A catalogue with a lower, better-characterised Mc for the TVZ specifically reduces the need for this kind of proxy correction in this particular source region, in principle allowing the smoothed-seismicity component of the national model to rely more directly on observed earthquake data here.

The Gutenberg-Richter b-value is the parameter through which this kind of catalogue improvement would most directly affect a hazard or catastrophe model's output, because it sets the slope of the magnitude-frequency relationship used to extrapolate from the observed population of smaller earthquakes to the much rarer, larger events that dominate loss estimates. A lower b-value implies a relatively higher rate of large events for a given rate of small events, a higher b-value implies the opposite. The b-values estimated in this project

varied substantially by catalogue, from 0.729 ± 0.177 (QuakeMigrate, bootstrap standard deviation) to 1.474 ± 0.073 (EQTransformer) and 1.594 ± 0.103 (PhaseNet), against a benchmark catalogue value of 0.908 over the same window. This is a materially wide range, illustrated in Figure 15. Taken at face value, and holding the observed small-event rate fixed, QuakeMigrate's b-value would imply a substantially higher extrapolated rate of larger, damaging earthquakes in the TVZ than either EQTransformer's or PhaseNet's own b-value would imply. Given QuakeMigrate's own small sample size (31 events above Mc) and correspondingly wide uncertainty, this specific numerical comparison should be treated as illustrative of the sensitivity involved rather than as a precise, competing estimate of the TVZ's true b-value. It nonetheless demonstrates, with specific values rather than a general assertion, exactly the kind of exceedance-curve sensitivity that motivates careful, tool-aware seismicity rate characterisation. The choice of detection method alone, independent of any genuine change in underlying seismicity, is capable of shifting a source zone's estimated large-event exceedance rate by a policy-relevant amount.

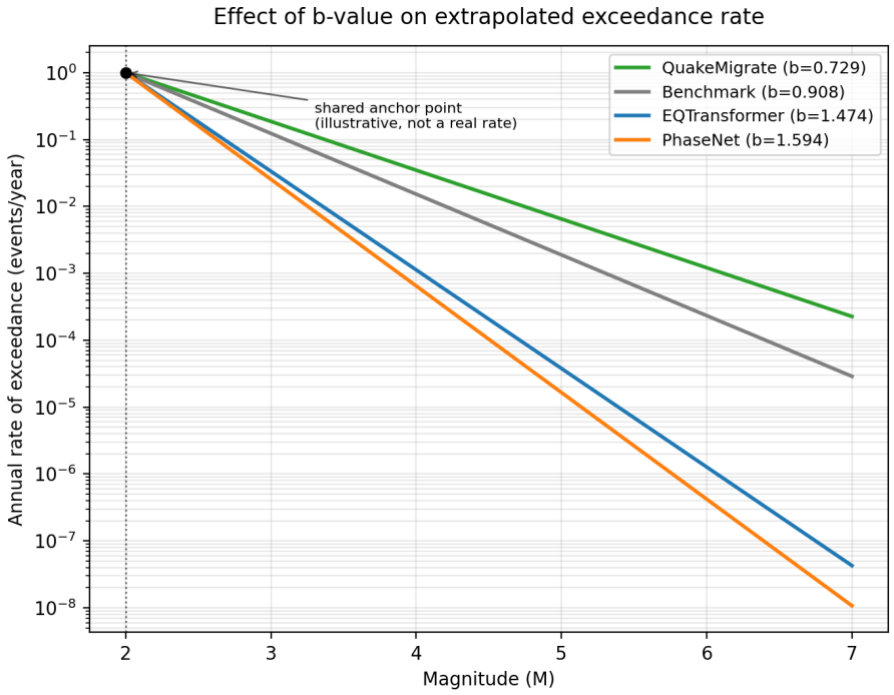


Figure 15: Illustrative effect of this project's catalogue-specific b-values on extrapolated exceedance rate, anchored to a shared hypothetical rate at M=2.0 to isolate the effect of b-value alone. Not a genuine multi-catalogue hazard forecast - catalogue-specific differences in Mc and completeness are not represented here.

The kernel density and difference maps produced in this project (Figures 6-11) make this sensitivity spatially explicit rather than only reporting a single zone-wide statistic. The difference map in particular (Figure 11) shows concretely where, within the TVZ, the choice of

detection tool would materially change the inferred spatial density of seismicity, which is directly what would need to be re-parameterised in a source-zone update to a probabilistic seismic hazard model. This matches the applied motivation set out in the original project proposal, that difference maps of this kind are directly relevant to how source-zone seismicity rates are parameterised in a PSHA or catastrophe model update.

The TVZ is not treated as a generic background source region within the New Zealand National Seismic Hazard Model, it already receives specific, dedicated attention. Independent testing of the 2022 revision found the TVZ to be one of only two notable regions nationally, alongside the central Alpine Fault, where a segmented fault-rupture model and a multi-fault model produce meaningfully different hazard estimates (Stirling et al., 2024). The ground-motion component of the model also applies a volcanic path attenuation term specific to the TVZ, added because ground-motion models not accounting for it were found to systematically over-predict shaking to the west of the zone. This existing, specific treatment of the TVZ within the national model supports the relevance of TVZ-specific catalogue improvements of the kind this project provides.

Verisk AIR's publicly available documentation of its New Zealand earthquake model places more emphasis on fault-specific rupture modelling, offering both a time-independent view (no memory of past ruptures) and a time-dependent view informed by known rupture history for specific mapped faults, and describes extensive peer review of the model's hazard component by local seismologists and academic researchers (AIR Worldwide, 2019). Its publicly available material provides less specific detail on background, off-fault seismicity-rate methodology than the GNS22 Seismicity Rate Model documentation. The general argument nonetheless applies regardless of a given model's specific, proprietary implementation, any earthquake catastrophe model ultimately depends on some form of Gutenberg-Richter-based seismicity-rate characterisation to extrapolate from observed, generally small-magnitude data to the rare, large events that drive loss, and a more complete, lower-Mc regional catalogue improves the observational basis of that characterisation for any model built on it.

4.7 Limitations

The single largest limitation of this project is its greatly reduced temporal scope. A two-month study window against the seventeen-year window originally proposed, necessitated by the computational complications with the High-performance Analytics and Learning Infrastructure (HALI) cluster (Section 1.4). This has two direct consequences: statistical estimates dependent on longer-term seismicity, particularly the Gutenberg-Richter b-value, are based on a comparatively small sample, especially for QuakeMigrate (31 events above Mc); and the study

window itself was deliberately chosen to encompass a known unrest episode, meaning this project's own results characterise an elevated-activity period rather than long-term background seismicity, a distinction discussed directly in Section 4.5.

The one-dimensional velocity model used throughout this project, in place of the fully three-dimensional structure the original study design specified (Section 2.4), carries a depth-accuracy cost, discussed in full in Section 4.2. QuakeMigrate's small final sample size (31 events) means its own Mc and b-value estimates in particular carry wide uncertainty (Table 3), and any numerical comparison drawn from them, such as the b-value comparison discussed in Section 4.6, should be treated as illustrative of a real sensitivity rather than a precise, competing estimate.

Local magnitude estimates throughout this project used a documented simplification of the published Taupō-specific attenuation relationship (Illsley-Kemp et al., 2021), setting the station correction term to zero rather than using the full published calibration table, which was not available from the published source (Section 2.7). Finally, the KDE bandwidth used throughout Sections 3.9-3.11, each catalogue's own median horizontal location uncertainty, is a principled but ultimately subjective methodological choice. A bandwidth sensitivity test, comparing the density and difference maps produced here against an alternative, independently justified bandwidth choice, was identified as a valuable check during this project's own risk planning but could not be completed within the schedule available, and is proposed as a specific piece of future work (Section 5).

5. Conclusions

This project set out to benchmark three automated earthquake detection and location methods, EQTransformer/PyOcto, PhaseNet/GaMMA and QuakeMigrate, against one another and against an independent benchmark catalogue, within the Taupō Volcanic Zone.

On RQ1, the workflow of Illsley-Kemp and Mestel (2025) was partially, not fully, reproduced. EQTransformer recovered 88% of the benchmark catalogue's event count and matched 53.7% of individual benchmark events, a substantial but incomplete overlap attributable to three specific, documented velocity-model and relocation-method simplifications. On RQ2, PhaseNet and QuakeMigrate were both benchmarked directly against EQTransformer within the TVZ. PhaseNet achieved closely comparable precision and recall to EQTransformer despite an entirely independent picker and associator, while producing more than double the raw pick volume, indicating that shared downstream methodology, not picker choice, was the dominant factor limiting both catalogues' final performance. QuakeMigrate achieved substantially higher precision but far lower recall, a direct consequence of the conservative

detection threshold required to keep its network-migration approach computationally tractable and free of noise-driven false detections in this volcanically active setting. On RQ3, this project's results connect to operational hazard characterisation: the magnitudes of completeness achieved by all three catalogues were substantially lower than GeoNet's routine detection threshold; and the resulting spread in estimated Gutenberg-Richter b-values was shown to be capable of shifting an extrapolated large-event exceedance rate for the TVZ by a policy-relevant amount, a finding directly relevant to the seismicity-rate component of the national seismic hazard model that at least one major catastrophe modelling provider explicitly incorporates.

Beyond these direct answers, this project's strongest and most robust findings are its convergent ones: a systematic depth-degradation pattern found independently in all three structurally different catalogues; and azimuthal gap as the dominant quality-filter rejection criterion in all three, both attributable to shared, physical limitations, the velocity model and the station network geometry, rather than to any tool's own algorithm. The concentration of activity around the September 4th 2019 mainshock was also found consistently across all three catalogues, and is well supported by independent, published evidence that this period was already part of a longer, documented volcanic unrest sequence.

Three pieces of future work follow directly from this project's own findings and limitations: extending the study period toward the originally proposed seventeen-year window, once computational resources allow, to properly characterise long-term background seismicity rather than an elevated-activity episode; replacing QuakeMigrate's static detection threshold with a dynamic, adaptive alternative, for example one based on the coalescence trace's own median absolute deviation, to address the noise-resonance limitation identified in Section 4.4; and conducting a systematic bandwidth sensitivity test for the kernel density estimates produced in Section 3.9, to establish how sensitive this project's spatial density and difference-map findings are to the specific bandwidth choice used throughout.

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